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Comparing Human and LLM Annotation of Media Frames in the ChangeMyView Dataset

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ABSTRACT

Although Large Language Models (LLMs) are highly efficient at processing structured formal data, their ability to interpret informal human communication remains a major challenge. Understanding this limitation is crucial if we want to use LLMs to speed up the analysis of online discussions. In current research, there are two big gaps. First, most research that tests models on framing tasks uses formal text as datasets, making it unclear how well LLMs handle finding frames in informal and persuasive discussions, like those on the ChangeMyView (CMV) subreddit. Second, while studies show that AI advice can bias human annotators on factual datasets, they miss what happens when humans get LLM suggestions for subjective tasks. This is crucial to see whether humans become overly dependent on LLM suggestions, which affects maintaining the quality and reliability of the dataset.

This research fills these gaps by testing both stand-alone models and hybrid human-LLM teams on the CMV dataset. To do this, a dataset of 495 unique comments was sampled from CMV. A human gold standard was created using Card et al.'s (2015) 15 framing dimensions. Different models were tested, to see how well they can perform the task. Finally, a user study looked at how humans and LLMs work together. It compared how humans annotate on their own against how they perform when they get LLM suggestions, allowing us to measure automation bias.

Our results show that while humans remain highly effective at interpreting implicit meaning, LLMs consistently struggle with these nuances. Furthermore, the hybrid annotation workflow failed to improve performance. Instead, it led to automation bias, which caused f1-scores to drop whenever participants accepted wrong LLM suggestions. This study concludes that without critical human oversight, current human-LLM collaboration can actually lower the quality instead of improving it. Because of this, human annotators are still necessary for framing tasks, and hybrid workflows need to be set up in a way that encourages annotators to stay critical of the LLM suggestions.

CONTENTS

PREFACE

After months of annotation, model training and writing, this thesis is now finished. I wrote this thesis as part of the bachelor program Information Science at the University of Groningen. During my studies, I have seen that LLMs have become much better at different tasks, for example, answering more complex questions. Because of the subjectivity of the task, I was excited to test whether LLMs were already better at the framing task than humans. Beforehand, I expected that the models would do a better job at the task than they actually did. It shows that there is still a lot of room for improvement when it comes to framing tasks. When looking at the final results, I noticed something else that I became curious about. I thought the participants in the experiment would follow the LLMs suggestions even more. I am curious why that is, whether they think that they are better at the task than LLMS, or they just simply do not trust LLMs. This is definitely something to investigate further in future research, especially since it connects information science with fields like psychology.

I want to thank my thesis supervisor, Khalid Al-Khatib, for providing us with the dataset and giving relevant feedback. I also want to thank Floris and Julius, for the annotating, so we could create the gold standard as a group. Although the task was quite time consuming, I am happy that we pushed through and managed to achieve the results we have.

1 INTRODUCTION

In the last couple of years, the use of Large Language Models (LLMs), such as ChatGPT, has increased rapidly. There have been concerns about the reliability of the generated texts, and an increasing number of studies have been done researching the potential of using LLMs as members of hybrid teams (Richter and Richter, 2024). Despite their growing popularity, it remains unclear to what extent LLMs improve the quality and reliability of human work. Wilkens (2020) states that AI tends to be a more valuable tool in situations where continuous monitoring and information processing are needed, but it has shortcomings in fields with a high need for creative problem solving and communication to build trust.

Although LLMs demonstrate high efficiency in processing structured formal data, their ability to interpret informal human communication remains the subject of debate (Wilkens, 2020). Informal discussions often rely on implicit meanings and creative language use, which continue to present challenges for automated systems. Identifying "frames", defined as the specific perspectives used to present an argument, such as 'Economic' or 'Political' perspectives, requires a deep understanding of nuance, cultural context, and communicative intent (Card et al., 2015). For example, while the isolated word "money" might typically be labelled under 'Economic', its context within a full sentence alters the appropriate framing. In the sentence, "Giving people financial aid after a disaster helps them live a happier life," the correct frame shifts to 'Quality of Life'. Because of this, humans may still maintain an advantage over LLMs in frame annotation tasks. Understanding this distinction is important for evaluating whether LLMs can function reliably as stand-alone tools or whether they are more effective when used collaboratively with humans.

To investigate the issue of LLMs limitations in understanding nuanced human communication, the ChangeMyView (CMV) subreddit provides a suitable dataset. CMV is an online platform where Reddit users can start a discussion with a specific opinion and other user can give arguments against this opinion to prove the original post is untrue. CMV consists of highly persuasive, informal, and often emotionally charged discussions, in addition, the quality is being checked by moderators (Tan et al., 2016). These characteristics make the dataset particularly valuable for studying the ability of both humans and LLMs to identify frames in complex online discussions.

Understanding how LLMs handle framing is important for several reasons. First, annotation tasks are very time consuming. Shortening the time it takes to annotate means that there is more data available for researchers to analyse large amounts of online debates. Second, it can tell us whether humans become overly dependent on using LLM suggestions when performing the task. This is crucial to maintaining the quality and reliability of the dataset. To explore this further, human-LLM collaboration can work in two main ways, LLMs can do tasks completely on their own without any human help, or they can work together in a hybrid team. The next chapter will explain these different setups of human-LLM interaction.

This study examines the performance of humans and LLMs in the frame annotation task of the CMV dataset. In addition, this study investigates the influence of human-LLM collaboration on annotation performance. More specifically, it explores whether human annotators critically evaluate LLM suggestions or become biased toward accepting them.

This study seeks to answer the following research question:
"How accurate are LLMs compared to humans for the frame annotation task of the ChangeMyView dataset and what is the influence of humans and LLMs working together on the annotation task?"

Leading to the following sub-questions:

1. *"How accurate are humans in the frame annotation task of the ChangeMyView dataset?"*
2. *"How accurate are LLMs in the frame annotation task of the ChangeMyView dataset?"*
3. *"How do the results of the LLMs compare to the human results?"*
4. *"What is the influence of humans and LLMs working together on the results of the frame annotation task of the ChangeMyView dataset?"*

Our results show a massive gap between the models and humans, with even ChatGPT-4o falling heavily short of the human baseline. Also, while LLM suggestions made human annotators agree with each other more, it actually lowered their overall score because they often agreed on the wrong frames by copying the LLMs mistakes.

The remainder of this thesis is structured as follows. Chapter 2 presents the relevant background literature. Chapter 3 covers the collection of the dataset, the annotation tasks, answering sub-question 1, and the processing steps used. Chapter 4 describes the methods used to answer the rest of the sub-questions, while Chapter 5 details the results and discussion. Lastly, Chapter 6 provides the conclusion and outlines the limitations of this study. All data, notebooks and python scripts that are used for the analysis in this thesis is available in public GitHub repository through the link in Appendix A.1.

2 BACKGROUND

To understand how humans and LLMs handle framing on CMV, we need to look at two main areas. First, we need to look at hybrid annotation workflows, focussing on how LLMs compare to humans and how working together affects data quality. Second, we look at media framing and why informal, emotional language on platforms like CMV makes framing so difficult. Together, these sections show what we already know and point out the gaps that this study aims to fill.

2.1 HYBRID ANNOTATION WORKFLOWS

As LLMs are increasingly integrated in tasks that involve reasoning, classification, and text interpretation, more research has been conducted on their impact. For example, Belcavello et al. (2025) evaluates the impact of semi-automating FrameNet semantic annotation by comparing manual human effort, fully automatic machine results, and a hybrid "Machine plus Human" approach. They found that hybrid annotation increases frame diversity and maintains similar coverage to human-only settings while preserving interpretive depth. Although fully automatic annotation performed significantly worse across most metrics and the hybrid model only slightly reduced overall annotation time, the semi-automatic workflow was shown to be a sustainable method for large-scale dataset growth.

Highlighting the performance gap between general-purpose LLMs and task-specific models, Boitel et al. (2023) found that task-specific BERT architecture (DeBERTa-v3-Large) achieved a 95% accuracy for emotion recognition on the BAUM dataset compared to GPT-3.5's (Davinci) 82.8%. Although GPT-3.5 showed higher throughput and handled data imbalances better, BERT's specialized architecture still achieved a higher accuracy. Although the authors note that models like GPT-4 may narrow this gap, these current limitations show that stand-alone LLMs still struggle with complex, subjective text. That is why researchers still need hybrid, human-in-the-loop workflows.

A similar conclusion was found by Kunjar et al. (2025) when looking specifically at framing tasks. They tested LLMs on news coverage and found that while these LLMs are helpful they are still often outperformed by manual human coders or smaller, specialized model. This shows again that stand-alone LLMs struggle with complex tasks like finding frames, and proves why we still need humans or hybrid teams to get the best results.

Despite its advantages in scalability, hybrid annotation may have unintended consequences for annotator behaviour and data quality. While Belcavello et al. (2025) argue that hybrid annotation is a sustainable method for large-scale dataset growth, Schroeder et al. (2025) warn of the psychological and systematic effects of this approach on the human annotator. The researchers conclude that providing LLM-generated suggestions for subjective annotation tasks does not reduce the time spent on the task, but does significantly increase annotators' self-reported confidence and task understanding. Furthermore, they found that annotators strongly adopt these AI suggestions, which homogenizes the "ground truth" toward the LLMs baseline and significantly alters the label distribution compared to unassisted human work.

Beck et al. (2026) investigate the psychological factors and system designs that influence human performance during the dependent verification of AI-generated pre-annotations. Tasking 2,784 participants with auditing greenhouse gas emissions data, the researchers analysed how interface design and user disposition affect the tendency toward under-correction. The experiment proved that human-AI loops are highly vulnerable to automation bias when interfaces require higher physical workload to overwrite errors. Furthermore, while financial incentives and early experiences with AI accuracy failed to alter human behaviour, individual attitudes toward automation and the conceptual complexity of the data were the definitive factors shaping whether humans successfully corrected or adopted to flawed AI recommendations.

To optimize the trade-off between annotation speed and data quality, Wang et al. (2024) proposed the LAPRAS framework. This collaborative framework enhances text annotation efficiency by combining machine-generated labels with strategic human verification. In this multi-step process, LLMs generate both labels and natural language explanations, which a separate verifier model then assesses to identify a subset of potentially incorrect annotations for human review. Experimental results across several benchmarks show that this method can improve overall dataset accuracy by at least 10% while requiring humans to re-annotate only 15% of the data. While human-subject studies showed smaller but still substantial gains around 7-8%. Further human studies indicate that while LLM-generated explanations assist in understanding decisions, their effectiveness in the re-annotation phase depends significantly on their quality and the specific task context.

2.2 MEDIA FRAMING ON SOCIAL PLATFORMS

An important question is whether these performance differences extend in other annotation tasks, such as the categorization of media frames. Card et al. (2015) introduces the 15 framing dimensions that we will use in this research. They define framing as the specific aspect of a topic that an argument emphasizes. The same topic can be discussed from different angles. The chosen angle is the frame, and it influences how audiences interpret the message.

This challenge is further amplified by the nature of the text being analysed. While previous research has been done on reliable and formal text. Belcavello et al. (2025), for example, used sentences from a corpus of television news media in Brazilian Portuguese. On the other hand, the CMV Reddit page is a page where people want to have discussions and maybe change their view on a specific topics. A foundational study by Tan et al. (2016) shows that persuasion on this platform relies heavily on the back-and-forth conversation and the words people choose. Most people do not use formal language in their comments and write their own unfiltered opinions. This makes the data highly emotional in nature, domains where, as Wilkens (2020) notes, AI tools often show shortcomings due to a lack of emotional reflection and social intelligence.

2.3 RESEARCH GAP

There are two big gaps in the current research. First, most paper that test models on framing tasks use formal news media as dataset (Belcavello et al., 2025; Kunjar et al., 2025). It is still unclear how well LLMs handle finding frames in informal and persuasive discussions, like those on the CMV subreddit. Second, while studies show that AI advice can bias human annotators (Schroeder et al., 2025; Beck et al., 2026), they mostly look at factual datasets, and miss what happens when humans get LLM suggestions for emotional and subjective tasks. This research fills these gaps by testing both models and human-LLM teams on the CMV dataset.

3 DATA AND MATERIAL

This chapter describes the collection of the dataset, the annotation, and processing steps used in this study. It begins by explaining the data collection process, including the source dataset from the ChangeMyView (CMV) subreddit and the strategy used for comment selection and sampling. Next, it introduces the annotation framework, including the 15 frame definitions, the labeling guidelines, the tool used for this task, and the inter-annotator agreement. Then Finally, this chapter outlines the processing pipeline, including data cleaning, gold standard construction, and prompt engineering for the LLM.

3.1 COLLECTION

This section explains the data collection process, including the source dataset from the CMV subreddit, as well as the strategy used for the comment selection and sampling. Together, these steps outline how the dataset of 500 comments were gathered before moving into the annotation phase.

3.1.1 Source dataset

The online discussions came from the sub-Reddit page ChangeMyView (CMV)[1]. The thesis supervisor provided us with a dataset with all the discussions and their meta data from the year 2005 to 2017, the Webis-CMV-20 dataset[2]. For this research, only the discussions are used and none of the meta data. From this dataset the thesis supervisor provided us with a sub-set of 50 discussions from the Webis-CMV-20 dataset. To make sure this data is high quality and covers different topics, these threads were filtered on comment richness, post completeness, topic uniqueness, and a balance across the years 2013 to 2017. All the specific specific sampling filters, engagement metrics, and structured statistics are in Appendix A.2.

Regarding the structure, each discussion consists of an 'Original Post (OP)', the post that started the discussion. Each OP has people reacting on it, these are the comments. For example, one OP can have one comment or can have over 50 comments.

3.1.2 Comment selection and sampling

Out of these 50 discussions a sub-set of 500 comments had to be chosen to use as final dataset, as requested by our thesis supervisor. For the first 200 comments, a sample from a parallel student research project was used because of an opportunity to compare the different annotations in future research. These comments are the total 200 first level comments of all threads.

To expand this dataset to a total of 500 comments, another 300 comments needed to be selected from the remaining data within the same 50 discussions. Instead of choosing these 300 comments manually, an LLM was used to randomly sample them. Using an LLM was the best option for two reasons. First, letting a human pick the comments takes more time and introduces selection bias because humans might unconsciously pick shorter or simpler comments. Using an LLM ensures a

[1] https://www.reddit.com/r/changemyview/
[2] https://zenodo.org/records/3778298

Frame	Description
Economic	costs, benefits, or other financial implications
Capacity and resources	availability of physical, human or financial resources, and capacity of current systems
Morality	religious or ethical implications
Fairness and equality	balance or distribution of rights, responsibilities, and resources
Legality, constitutionality and jurisprudence	rights, freedoms, and authority of individuals, corporations, and government
Policy prescription and evaluation	discussion of specific policies aimed at addressing problems
Crime and punishment	effectiveness and implications of laws and their enforcement
Security and defense	threats to welfare of the individual, community, or nation
Health and safety	health care, sanitation, public safety
Quality of life	threats and opportunities for the individual's wealth, happiness, and well-being
Cultural identity	traditions, customs, or values of a social group in relation to a policy issue
Public opinion	attitudes and opinions of the general public, including polling and demographics
Political	considerations related to politics and politicians, including lobbying, elections, and attempts to sway voters
External regulation and reputation	international reputation or foreign policy of the U.S.
Other	any coherent group of frames not covered by the above categories

Table 3.1: The 15 different annotation frames of Card et al. (2015).

completely randomized and unbiased selection from the remaining threads. Second, the LLM could write and execute the data selection code very quickly. The prompt used to select the last 300 comments can be found in Appendix A.3.

3.2 ANNOTATION

This section describes the set of 15 argumentative frames used to capture different perspectives, as well as the labeling guidelines applied during annotation. These guidelines define how frames should be identified and how multiple frames within a single comment are handled. Together, the framework and guidelines ensure a consistent and structured approach to assigning labels across the dataset. Label Studio [3], an open-source annotation tool, is used to organize and annotate the dataset. Additionally, this section presents the inter-annotator agreement scores and explains the adjudication process used to create the gold standard.

3.2.1 The labels

The annotation task is based on 15 distinct dimensions, or 'frames'. These frames represent different perspectives from which a topic can be argued (Card et al., 2015). Table 3.1 provides a detailed overview of each frame, and its definition.

[3] https://labelstud.io/

CMV Comment	Frame
“CMV: Religion has no positive attributes”	morality
“CMV: Money to me is a renewable source so I just spend it knowing I’ll get paid again so it’ll come back”	economic
“CMV: There are no adequate gender equal options when it comes to changing your surname in marriage”	fairness and equality
“CMV: Watches today serve no functional purpose other than fashion, and therefore are a waste of money”	public opinion
“CMV: Most "big words" have no place outside of formal writing or speeches.”	other

Table 3.2: Examples of CMV titles and their corresponding annotation labels.

CMV Comment	Frames
“CMV: I believe that blacks are genetically inferior to whites, Asians, and jews as far as intelligence is concerned.”	‘morality’ and/or ‘cultural identity’

Table 3.3: Example of a CMV title were multi-frame labeling is possible.

3.2.2 Annotation examples

To better understand the annotation process, consider how context changes a text’s classification. For instance, while the isolated word "discrimination" might typically be labelled under ’Morality’, its context within a full sentence alters the appropriate framing. In the sentence, "CMV: Discrimination is discrimination no matter who it’s against," the correct frames shift to ’Public Opinion’ and ’Cultural Identity’. Similarly, a topic like "marriage" could easily look like ’Cultural Identity’, but the sentence "CMV: There are no adequate gender equal options when it comes to changing your surname in marriage" focuses on ’Fairness and Equality’. To further clarify the application of these labels, table 3.2 presents more specific examples of the CMV dataset and their assigned frames.

3.2.3 Handling Multi-Frame Examples

When considering two or more equally valid frames, choose all. This is good for reducing subjectivity. In other words, multi-frame issues are possible. An example of when multi-frame labeling is possible is seen in table 3.3.

3.2.4 Significant findings during annotation

During the annotation process we made several notable observations. First, the LLM never suggested the ’Other’ frame, even in contexts where it would be appropriate. Instead, the LLM occasionally provided no suggestion at all, even though the ’Other’ frame was highly appropriate given the context of the comment. Secondly, the LLM frequently over-suggested the ’Economic’ frame, although it was incorrect. In those cases, the LLM was probably triggered by the word "money" or some kind of number that appeared in the comment. Lastly, the ’External Regulation and Reputation’ frame was rarely used because it was only suitable for a few comments.

3.2.5 Annotation tasks

For the annotation, Label Studio was used. This provided a tool so every annotator could do the annotations at the same time and on their own device. For every comment that needed to be annotated, the OP and the comments were shown.

All 500 comments have LLM suggestions, the human annotators can agree with the suggestions or choose the label(s) they see fit best. The previously mentioned first 200 comments are annotated by three humans. These 200 comments are split into 150 comments annotated first, and 50 annotated after. Guidelines were refined after the 150 and 50 comments. Finally, each human independently annotated 100 comments, summing up to 500 comments in total.

There were three annotators, they are labelled annotator 1, 2, and 4 by Label Studio. We kept this labeling consistent through this thesis and all relevant documents.

3.2.6 Evaluation metrics

To evaluate the overall agreement between the annotators across the 15 different frames after the annotation of the first 200 comments, Krippendorff's Alpha is used (Krippendorff, 2018). Standard metrics, like Cohen's Kappa or standard nominal metrics, only work when a comment has exactly one label. However, as explained in section 3.2.3, a comment can have more than one frame. Because of these multi-label data, standard metrics are not suitable because they would count a partial agreement as a complete disagreement. Therefore, Krippendorff's Alpha with MASI (Passonneau, 2006) and Jaccard distance (Jaccard, 1912) were chosen because these metrics are designed to handle partial overlap. While Jaccard measures strict mathematical overlap, MASI is used because it down-weights penalties for subset relationships. In addition to Alpha, the mean pairwise Jaccard score is also calculated to show the average agreement between specific pairs of annotators. Finally, while Alpha scores are used to see the overall agreement, Fleiss's Kappa (Fleiss, 1971) is used to track the agreement for each individual frame.

Agreement on the set of 200 triple-annotated comments

To check annotator alignment, inter-annotator agreement was calculated in three stages: the first 150 comments, the next 50 comments, and the combined set of 200 comments. The Krippendorff's Alpha MASI and Jaccard scores are shown in table 3.4. The Jaccard pairwise results are shown in table 3.5, and the individual frame agreements are shown in table 3.6.

For the first 150 comments, Alpha MASI was 0.4448 and Jaccard was 0.5089, with 'Economic' and 'Political' frames scoring highest. For the next 50 comments, Alpha dropped to 0.4044, but the worst-performing frames improved significantly. For example, 'External Regulation and Reputation' improving from -0.01 to 0.52. This shows that the refined guidelines helped. Combined for all 200 comments, Alpha MASI was 0.4356 and Jaccard was 0.5093, these low scores were expected given the 15 different frames. Across all categories, the Fleiss' Kappa scores ranged from 0.34 to 0.79.

Dataset	MASI	Jaccard
First 150 Comments	0.4448	0.5089
Last 50 Comments	0.4044	0.5045
All 200 Comments	0.4356	0.5093

Table 3.4: Inter-annotator agreement measured using Krippendorff's Alpha with MASI and Jaccard distance.

Last 300 single-annotated comments

To scale the dataset, each of the three annotators independently annotated their own set of 100 comments. They use the refined annotation guidelines found in Appendix A.4. Because these comments were only annotated by one person, no

Pair	First 150	Last 50	Total 200
1-2	0.5940	0.6900	0.6180
1-4	0.5858	0.5443	0.5754
2-4	0.5096	0.4950	0.5059
Mean	**0.5631**	**0.5764**	**0.5664**

Table 3.5: Summary of mean pairwise Jaccard scores across annotation sets.

Frame	First 150	Last 50	Total 200
Economic	0.7733	0.7847	0.7859
Political	0.6956	0.8010	0.7216
Crime and Punishment	0.6793	0.7432	0.6897
Morality	0.6691	0.4996	0.6238
Legality, Constitutionality	0.5993	0.6296	0.6064
Health and Safety	0.6199	0.5273	0.5985
Fairness and Equality	0.5377	0.6054	0.5597
Security and Defense	0.5909	0.4947	0.5479
Cultural Identity	0.5183	0.5694	0.5387
Quality of Life	0.5105	0.4883	0.5122
Public Opinion	0.4733	0.5356	0.4877
Policy Prescription	0.4311	0.4029	0.4253
Capacity and Resources	0.3944	0.1319	0.3538
External Regulation	-0.0112	0.5272	0.3418
Other	0.4915	0.6019	0.5211

Table 3.6: Summary of Fleiss' Kappa scores by frame across annotation sets.

inter annotator agreement is calculated, and the annotations directly serve as the gold standard.

3.2.7 Evaluation

To answer the question "How accurate are humans in the frame annotation task of the ChangeMyView dataset?", the accuracy, precision, recall and f1-scores are calculated per annotator to see how well the annotators score against the gold standard. The overall accuracy scores are between 0.520 and 0.725. However, when looking at the accuracy per frame for each annotator in tables A.1, A.2 and A.3, the accuracy lies between 0.895 and 0.995. It is to be expected that the scores per frame are higher because each frame is looked at individually, meaning there are many correct matches where a frame is simply not present. The overall precision is between 0.7429 and 0.8625, the overall recall is between 0.7958 and 0.8879, and the overall f1-score is between 0.7458 and 0.8658. Most of these scores are considered good, meaning they are above 0.7. Only the accuracy scores of annotator 2 and 4 are below that line of 0.7 which means that they still are quite accurate but less accurate than annotator 1.

It is also important to remember that the gold standard was made by the same annotators through adjudication. Because of this, comparing their initial annotations against this gold standard does not measure their accuracy against an outside truth. Instead, it shows how close each annotator's own choices were to what the group eventually agreed on. For example, the high scores of annotator 1 show that their independent choices were already very close to the gold standard the group agreed on.

The reason why the overall accuracy scores are lower than the f1-scores is because accuracy is very strict for multi-label data. For the overall accuracy score to count as match, the annotator must choose the exact same combination of frames as the gold standard. If the annotator misses at least one of the frames, the accuracy score for that comment is zero, even though the precision and recall remain high. This shows that the annotators were very good at identifying the correct frames, but matching the exact combination is difficult because of the many frames to choose from.

To answer the question, humans are overall highly accurate at this framing task and get better at the task over time. Even though overall accuracy is lower for some annotators, the high f1-scores (all above 0.74) prove that human annotators can reliably identify frames in the CMV comments.

Annotator	Accuracy	Precision	Recall	f1
1	0.725	0.8625	0.8879	0.8658
2	0.560	0.8192	0.8408	0.8092
4	0.520	0.7429	0.7796	0.7458

Table 3.7: Classification performance per annotator pre-adjudication.

3.3 PROCESSING

This section explains the processing pipeline, including data cleaning, gold standard construction, as well as post-adjudication agreement and evaluation. Additionally, it outlines the prompt engineering used to generate the LLM suggestions. Together, these steps ensure that the dataset is both reliable and well-suited for comparing human and LLM performance on the frame annotation task.

3.3.1 Cleaning

While establishing the gold standard, five duplicate comments were identified in the dataset. These comments can be found in table A.4. Therefore, these duplicates needed to be removed. These duplicates remained in the first 200 triple-annoated comments but were removed from the last 300 single-annotated comments. Reducing the dataset to 495 comments.

3.3.2 Gold standard construction

The Gold Standard is established through an adjudication process. After the independent annotation, the overlapping frames are compared. In cases of disagreement, the labels that were chosen by at least two of the three annotators are selected for the gold standard. Comments without agreement are discussed, and the best labels are chosen in mutual agreement. The comments that are annotated by only one human, immediately count as ground truth. The finalized set of 495 comments serves as the ground truth for training and testing the models.

3.3.3 Agreement Post-adjudication

To check annotator alignment after resolving disagreements, the post-adjudication inter-annotator agreement was calculated against the same three stages. The Krippendorff's Alpha MASI and Jaccard scores are shown in table 3.8. The combined Jaccard results are shown in table 3.9, and the individual frame agreements are shown in table 3.10.

For the first 150 comments, the overall Alpha MASI was 0.4666 and Jaccard was 0.5313, with 'Economic' and 'Political' frames scoring the highest. For the next 50 comments, Alpha MASI dropped slightly to 0.4267 and Jaccard to 0.5280, but the lower-performing frames showed improvement. Combined for all 200 comments, Alpha MASI was 0.4575 and Jaccard was 0.5318. 'Economic' (0.7859) and 'Political' (0.7444) remained the most reliable frames, while final Fleiss' Kappa scores ranged from 0.3418 to 0.7859 across all categories.

Dataset	MASI	Jaccard
First 150 Comments	0.4666	0.5313
Last 50 Comments	0.4267	0.5280
All 200 Comments	0.4575	0.5318

Table 3.8: Post-adjudication inter-annotator agreement measured using Krippendorff's Alpha with MASI and Jaccard distance metrics.

Pair	First 150	Last 50	Total 200
1-2	0.6140	0.7100	0.6380
1-4	0.6058	0.5643	0.5954
2-4	0.5296	0.5150	0.5259
Mean	**0.5831**	**0.5964**	**0.5864**

Table 3.9: Summary of mean pairwise Jaccard scores post-adjudication.

Frame	First 150	Last 50	Total 200
Economic	0.7733	0.7847	0.7859
Political	0.7264	0.8010	0.7444
Crime and Punishment	0.6793	0.7432	0.6897
Morality	0.6846	0.4996	0.6349
Legality, Constitutionality and Jurisprudence	0.5993	0.6296	0.6064
Health and Safety	0.6199	0.5273	0.5985
Fairness and Equality	0.5716	0.6054	0.5833
Security and Defense	0.5909	0.4947	0.5479
Quality of Life	0.5638	0.4883	0.5450
Public Opinion	0.5386	0.5356	0.5385
Cultural Identity	0.5364	0.5694	0.5519
Other	0.5214	0.6502	0.5556
Capacity and Resources	0.4591	0.4718	0.4624
Policy Prescription and Evaluation	0.4423	0.5179	0.4606
External Regulation and Reputation	-0.0112	0.5272	0.3418

Table 3.10: Summary of Fleiss' Kappa scores by frame post-adjudication.

3.3.4 Evaluation post-adjudication

When comparing the scores before and after adjudication, it is clear that the performance of all annotators improved. The scores per frame for each annotator can be found in tables A.5, A.6 and A.7. The overall accuracy for each annotator increased by exactly 0.02, while precision, recall, and f1-scores all saw small improvements. This general improvement is to be expected because both calculations are compared against the same final gold standard. In the post-adjudication evaluation, the final annotations are used, which were updated after discussing disagreements. Because of this, the annotators' final choices naturally match the gold standard better than their initial choices. This shows that the adjudication process worked well to bring the individual annotators closer to the final agreement.

Annotator	Accuracy	Precision	Recall	f1
1	0.745	0.8725	0.9029	0.8791
2	0.580	0.8217	0.8521	0.8181
4	0.540	0.7554	0.7958	0.7608

Table 3.11: Classification performance per annotator post-adjudication.

3.3.5 Prompt engineering

A prompt was provided to the LLM to generate suggestions for human annotators for all 500 comments. For the first 200 comments no definition of the frames was given to the LLM. Adding the definition in the prompt for the last 300 comments was expected to give better results. The full annotation prompt is provided in Appendix A.8.

To see how well the LLM performed, the mean Jaccard score was calculated between the gold standard and the LLM suggestions. However, the score slightly decreased from 0.5797 to 0.5226 (table 3.12). This drop is likely because the last 300 comments were only annotated by a single annotator without an adjudication process, making that section of the gold standard less consistent. Another reason could be that the annotators got better at the annotation task and were more critical at the LLM suggestions.

Comment Set	Mean Jaccard
First 200	0.5797
Last 300	0.5226
All 500	0.5455

Table 3.12: Mean Jaccard scores between the gold standard and the LLM suggestions.

4 METHOD

This chapter describes the method used to evaluate LLMs for text annotation. It explains how the dataset was split, which models were chosen, and how they were trained and evaluated. Next, it describes how LLM performances are compared to human performance. Finally, it describes the user study experiment that tests whether LLM suggestions cause automation bias in human annotators.

4.1 HOW ACCURATE ARE LLMS IN THE FRAME ANNOTATION TASK OF THE CHANGEMYVIEW DATASET?

4.1.1 Distribution of training & testing set

The dataset of 495 comments was split in a 80% training/validation set (369 comments) and a 20% test set (99 comments). To tune the hyper-parameters and prevent over-fitting, the training set is split again in a train set (322 comments) and a validation set (74 comments). To make sure the experiments are reproducible, the data was split using a fixed random seed (random state = 42). Because labeling the 15 different frames is a multi-label classification task, a single comment can have multiple overlapping frames, the text labels were converted into binary vectors before being fed into the models.

4.1.2 Model selection

Five different models were chosen across three categories, encoders, decoders, and API-based LLM. To evaluate their performance on the frame annotation task.

- **BERT base uncased:** This is used as the baseline traditional encoder model. We added a custom linear layer on top to see how a smaller, task-specific model works when directly fine-tuned.
- **ModernBERT:** This is a newer and more optimized version of BERT. We wanted to see if a more modern encoder performs better than the baseline.
- **Qwen3.5-2B:** This smaller generative model was set up as a sequence classifier to evaluate how a small language model performs on this task.
- **Qwen3.5-0.8B:** This is an even smaller version of the same family as the 2B model. This lets us check how reducing the model size affects accuracy.
- **ChatGPT-4o:** This is a closed-source language model accessed via an OpenAI API. Instead of local fine-tuning, we used it to test few-shot-in-context learning.

4.1.3 Training

For fine-tuning the BERT, ModernBERT, and Qwen models, we had to deal with severe class imbalance and rare frames. To stop the models from just predicting zeros for everything, we modified the standard Hugging Face trainer into a custom WeightedTrainer with a modified loss function.

This trainer dynamically calculates positive weights based on how often a frame appears in the train set (322 comments). To maintain gradient stability, these scaling factors are clamped between 1.0 and 50.0 and passed directly to the BCEWithLogitsLoss function. In this way, the model is forced to focus its attention on capturing infrequent frames rather than defaulting to majority classes.

The training configurations for the models lasted for five epochs:

- **Sequence length:** 512 tokens for BERT and ModernBERT. 1024 tokens for Qwen models to fit more context with explicit end-of-sequence padding.
- **Batch sizes:** Total effective batch size of 16. BERT models used a batch size of 16 natively, while larger Qwen models used a batch size of 1 with 16-step gradient accumulation to prevent memory errors.
- **Optimisation:** All models use the AdamW optimizer with a weight decay of 0.01. BERT models used a learning rate of 2×10^{-5} using fp16 precision. The Qwen models used a learning rate of 4×10^{-5} and bf16 precision. The final checkpoints were selected using the highest validation micro f1-score.

As ChatGPT-4o was not fine-tuned, a six-shot approach was used. Placing 6 random training examples with gold-standard labels and the full definitions of all 15 frames into the system prompt. The temperature was set to 0.0 for consistency.

4.1.4 Tuning decision threshold

Instead of the default 0.5 threshold, we tuned it using the validation set. The script tested continuous probabilities from 0.10 to 0.70 with steps of 0.05, choosing the specific threshold that achieved the highest validation micro f1-score for final evaluation on the test set.

4.1.5 Evaluation

To evaluate model performance, accuracy, precision, recall and f1-score were calculated for each model against the human gold standard.

4.2 HOW DO THE RESULTS OF THE LLMS COMPARE TO THE HUMAN RESULTS?

The mean scores of the annotators will be calculated, pre- and post-adjudication. These scores will be compared to the scores of each of the LLMs.

4.3 WHAT IS THE INFLUENCE OF HUMANS AND LLMS WORKING TOGETHER ON THE RESULTS OF THE FRAME ANNOTATION TASK OF THE CHANGEMYVIEW DATASET?

4.3.1 The goal

The goal of the experiment is to test whether humans will follow the LLM suggestions when annotating frames or if they correct the wrong suggestions. The experiment will test two aspects, one being how good the participants are in the annotation task on their own. The other being how good the participants are in the annotation when given LLM suggestions. The study tests for critical thinking or incorrectly following the LLMs' errors.

4.3.2 Experimental setup

The experiment is hosted via Google Forms. The form will start with an introduction, the consent confirmation checkbox, and the annotation guidelines.

The guidelines given are a summarized version of the previous stated guidelines. Unlike the original guideline, participants can only choose one or two frames and can only choose one frame when choosing 'other' as the frame. This is to make the annotation process easier for people who have never annotated. The complete annotation guidelines stated in the form can be found in Appendix A.9.

In part A, per comment all the different frames with definition are stated underneath each other. The participants need to choose the frame they think suits the comments best.

Part B, is mostly the same as part A, however, a LLM suggestion is given. The LLM suggestions is given underneath the comment and will be given in the form "suggestion: 'frame(s) x'". 50% of the LLM suggestions are correct and 50% are incorrect. The participant do not know that the suggestions could be incorrect.

The first part must be completed first, after which the second part will be completed. For group A the first part is A, and for group B the first part is B. Parts B(i) and B(ii) are randomized, but the order of the comments is the same for all participants. The entire experiment can be found in the table A.8.

Twenty participants filled in the form. Ten participants completed first part A then part B, this will be group A. The other ten participants first completed part B then part A, this will be group B.

4.3.3 Evaluation

To evaluate the performance, human annotators from both parts were compared against the gold standard. Accuracy, precision, recall, and f1-scores will be calculated for each participant across different conditions. To measure automation bias directly, the exact copy rate is calculated, measuring how often participants accepted the LLM suggestion without making any modifications, broken down by whether the LLM was correct or incorrect.

5 RESULTS AND DISCUSSION

5.1 HOW ACCURATE ARE LLMS IN THE FRAME ANNOTATION TASK OF THE CHANGEMYVIEW DATASET?

5.1.1 BERT

BERT's loss weights caused a heavy precision-recall trade-off. With a 0.55 threshold, the model predicted 3.36 frames per comment on average, the baseline is 1.96, capturing rare frames but causing high false positives. Consequently, total precision was 22.82% and the accuracy was 3.0%.

"Political" and "Other" scored highest with an f1 score of 0.6667. In contrast, sparse categories had severe imbalances. "Capacity and Resources" reached 71.43% recall but just 16.13% precision, and rare classes like 'External Regulation and Reputation', 'Legality, Constitutionality and Jurisprudence', and 'Security and Defense' were completely missed, with f1-scores of 0.0000. The complete results are shown in table 5.1.

Frame	Accuracy	Precision	Recall	f1
Capacity and Resources	0.717	0.1613	0.7143	0.2632
Crime and Punishment	0.737	0.1481	0.5714	0.2353
Cultural Identity	0.667	0.3529	0.2143	0.2667
Economic	0.697	0.2000	0.5000	0.2857
External Regulation and Reputation	0.970	0.0000	0.0000	0.0000
Fairness and Equality	0.586	0.2195	0.5000	0.3051
Health and Safety	0.556	0.1489	0.6364	0.2414
Legality, Constitutionality and Jurisprudence	0.828	0.0000	0.0000	0.0000
Morality	0.596	0.3333	0.4815	0.3939
Policy Prescription and Evaluation	0.758	0.1538	0.1333	0.1429
Political	0.949	0.7143	0.6250	0.6667
Public Opinion	0.677	0.2083	0.2778	0.2381
Quality of Life	0.646	0.1944	0.5385	0.2857
Security and Defense	0.899	0.0000	0.0000	0.0000
Other	0.929	0.5833	0.7778	0.6667
Total for all frames	0.030	0.2282	0.3918	0.2884

Table 5.1: Classification performance for BERT.

5.1.2 ModernBERT

ModernBERT's results also reflect a heavy precision-recall trade-off caused by the positive loss weights. With a 0.5 threshold, the model predicted 3.56 frames per comment on average, capturing rare frames but causing massive false positives. Consequently, total precision was 35.23% and the accuracy fell to 2.0%.

'Health and Safety' and 'Political' performed best, with f1-scores of 0.6429 and 0.6087. In contrast, the sparse categories had severe imbalances. 'Capacity and Resources' hit 57.14% recall but just 19.05% precision. Rare classes were no longer completely missed. The complete results are shown in table 5.2.

Frame	Accuracy	Precision	Recall	f1
Capacity and Resources	0.798	0.1905	0.5714	0.2857
Crime and Punishment	0.848	0.2143	0.4286	0.2857
Cultural Identity	0.747	0.5429	0.6786	0.6032
Economic	0.788	0.3043	0.5833	0.4000
External Regulation and Reputation	0.980	1.0000	0.3333	0.5000
Fairness and Equality	0.687	0.3243	0.6667	0.4364
Health and Safety	0.899	0.5294	0.8182	0.6429
Legality, Constitutionality and Jurisprudence	0.677	0.2571	0.6000	0.3600
Morality	0.626	0.3864	0.6296	0.4789
Policy Prescription and Evaluation	0.768	0.3667	0.7333	0.4889
Political	0.909	0.4667	0.8750	0.6087
Public Opinion	0.707	0.3333	0.6111	0.4314
Quality of Life	0.768	0.2727	0.4615	0.3429
Security and Defense	0.939	0.2000	0.3333	0.2500
Other	0.848	0.3500	0.7778	0.4828
Total for all frames	0.020	0.3523	0.6392	0.4542

Table 5.2: Classification performance for ModernBERT.

5.1.3 Qwen3.5-2B

Qwen3.5-2b results show a much better balance between recall and precision, significantly reducing over-prediction. With a 0.55 threshold, the model only predicting 2.11 frames per comment on average. This minimized false positives while capturing rare frames, improving total precision to 51.20% and accuracy to 14.10%.

'Political' and 'Cultural Identity' performed best, with f1-scores of 0.7000 and 0.6316. However, custom weights still caused slight imbalances in sparse categories, 'Capacity and Resources' hit 57.14% recall but just 23.53% precision. Rare classes like 'Legality, Constitutionality and Jurisprudence' and 'Security an Defense' were successfully captured, though 'External Regulation and Reputation' was completely missed, with an f1-score of 0.0000. The complete results are shown in table 5.3.

Frame	Accuracy	Precision	Recall	f1
Capacity and Resources	0.838	0.2353	0.5714	0.3333
Crime and Punishment	0.939	0.6667	0.2857	0.4000
Cultural Identity	0.788	0.6207	0.6429	0.6316
Economic	0.859	0.4500	0.7500	0.5625
External Regulation and Reputation	0.970	0.0000	0.0000	0.0000
Fairness and Equality	0.859	0.6250	0.5556	0.5882
Health and Safety	0.899	0.5333	0.7273	0.6154
Legality, Constitutionality and Jurisprudence	0.838	0.4737	0.6000	0.5294
Morality	0.717	0.4828	0.5185	0.5000
Policy Prescription and Evaluation	0.879	0.6154	0.5333	0.5714
Political	0.939	0.5833	0.8750	0.7000
Public Opinion	0.788	0.4000	0.3333	0.3636
Quality of Life	0.879	0.5333	0.6154	0.5714
Security and Defense	0.980	1.0000	0.3333	0.5000
Other	0.919	0.6000	0.3333	0.4286
Total for all frames	0.141	0.5120	0.5515	0.5310

Table 5.3: Classification performance for Qwen2B.

5.1.4 Qwen3.5-0.8B

Qwen3.5-0.8b results also reflect a heavy precision-recall trade-off caused by the positive loss weights. With a low 0.30 threshold, the model predicted 3.51 frames per comment on average, capturing rare frames but generating massive false positives. Consequently, the total precision was 40.06% and the accuracy was 7.1%.

'Political' and 'Morality' performed best, with f1-scores of 0.8889 and 0.6154. In contrast, custom weights still caused severe imbalances in sparse categories, 'Capacity and Resources' hit 57.14% recall but just 28.57% precision. Rare classes like 'Legality, Constitutionality and Jurisprudence' and 'Security and Defense' were successfully captured, though, 'External Regulation and Reputation' was completely missed,with an f1-score of 0.0000. The complete results are shown in table 5.4.

Frame	Accuracy	Precision	Recall	f1
Capacity and Resources	0.869	0.2857	0.5714	0.3810
Crime and Punishment	0.919	0.4444	0.5714	0.5000
Cultural Identity	0.657	0.4444	0.8571	0.5854
Economic	0.828	0.3684	0.5833	0.4516
External Regulation and Reputation	0.970	0.0000	0.0000	0.0000
Fairness and Equality	0.778	0.4091	0.5000	0.4500
Health and Safety	0.848	0.4167	0.9091	0.5714
Legality, Constitutionality and Jurisprudence	0.727	0.3421	0.8667	0.4906
Morality	0.747	0.5263	0.7407	0.6154
Policy Prescription and Evaluation	0.828	0.4643	0.8667	0.6047
Political	0.980	0.8000	1.0000	0.8889
Public Opinion	0.596	0.2800	0.7778	0.4118
Quality of Life	0.697	0.2571	0.6923	0.3750
Security and Defense	0.970	0.5000	0.3333	0.4000
Other	0.929	0.7500	0.3333	0.4615
Total for all frames	0.071	0.4006	0.7165	0.5139

Table 5.4: Classification performance for Qweno.8B.

5.1.5 ChatGPT-4o

ChatGPT-4o results reflect a much better balance between recall and precision, relying on the strong pre-trained text understanding of an LLM. With a six-shot approach, the model over-assigned frames slightly, predicting 2.39 frames per comment on average. This successfully captured rare frames while minimizing false positives, allowing total precision to reach 58.23% and accuracy to rise to 17.20%.

'Health and Safety' and 'Economic' performed best, with f1-scores of 0.7500 and 0.7273. However, the in-context learning setup still caused some imbalances in sparse categories, 'Capacity and Resources' hit 57.14% recall but just 28.57% precision. Rare classes like 'External Regulation and Reputation', 'Legality, Constitutionality and Jurisprudence' and 'Security and Defense' were successfully captured, returning f1-scores of 0.3333, 0.6667, and 0.6000. The complete results are shown in table 5.5.

5.1.6 Summary of results

To summarise, the results show a trade-off between recall and precision across most tested models. The smaller fine-tuned models (BERT, ModernBERT, Qwen3.5-0.8B) over-predicted frames due to the positive loss weights in the WeightedTrainer. While this strategy successfully captured rare frames, it generated massive false positive and heavily lowered the accuracy. On the other side, a larger model or a prompted approach achieved a much better balance. Qwen3.5-2B reduced over-prediction by using a higher threshold, while ChatGPT-4o performed best overall, demonstrating that a six-shot prompting method relying on strong pre-trained text understanding provides well-balanced frame annotations without the severe false positives of local fine-tuning.

Frame	Accuracy	Precision	Recall	f1
Capacity and Resources	0.869	0.2857	0.5714	0.3810
Crime and Punishment	0.929	0.5000	0.7143	0.5882
Cultural Identity	0.818	0.6471	0.7857	0.7097
Economic	0.909	0.5714	1.0000	0.7273
External Regulation and Reputation	0.960	0.3333	0.3333	0.3333
Fairness and Equality	0.899	0.8333	0.5556	0.6667
Health and Safety	0.939	0.6923	0.8182	0.7500
Legality, Constitutionality and Jurisprudence	0.879	0.5714	0.8000	0.6667
Morality	0.838	0.7895	0.5556	0.6522
Policy Prescription and Evaluation	0.869	0.5556	0.6667	0.6061
Political	0.879	0.4000	1.0000	0.5714
Public Opinion	0.869	0.6000	0.8333	0.6977
Quality of Life	0.899	0.5789	0.8462	0.6875
Security and Defense	0.960	0.4286	1.0000	0.6000
Other	0.919	1.0000	0.1111	0.2000
Total for all frames	0.172	0.5823	0.7113	0.6404

Table 5.5: Classification performance for ChatGPT-4o.

5.2 HOW DO THE RESULTS OF THE LLMS COMPARE TO THE HUMAN RESULTS?

Model	Accuracy	Precision	Recall	f1
Mean annotators (pre-adjudication)	0.6017	0.8082	0.8361	0.8069
Mean annotators (post-adjudication)	0.6217	0.8165	0.8503	0.8193
BERT	0.030	0.2282	0.3918	0.2884
ModernBERT	0.020	0.3523	0.6392	0.4542
Qwen3.5-2B	0.141	0.5120	0.5515	0.5310
Qwen3.5-0.8B	0.071	0.4006	0.7165	0.5139
ChatGPT-4o	0.172	0.5823	0.7113	0.6404

Table 5.6: Mean performance metrics across annotators and models.

When comparing the LLMs to the human results, there is a massive gap in overall performance (table 5.6). Human annotators performed incredibly well, reaching an initial pre-adjudication f1-score of 80.69% and 60.17% accuracy. After adjudication resolved disagreements, all numbers improved, bringing precision to 81.65%, recall to 85.03%, and accuracy to 62.17%. However, because these same annotators created the gold standard, comparing their initial choices against it shows how close they were to the final group agreement, rather than measuring accuracy against an outside truth.

In contrast, the models were tested against this gold standard as completely external evaluators. Even the best model, ChatGPT-4o, falls heavily short of the human baseline, falling short of the post-adjudication score by nearly 18% with an accuracy of just 17.20%. Fine-tuned models score even lower. For example, BERT's accuracy drops to 3.0%. Also, while humans balanced precision and recall, the models show a heavy imbalance due to their over-prediction.

Per-frame human performance in the appendix (tables A.10 and A.11) shows which frames were easier or harder to define. The 'Economic' frame was the easiest for humans, with a strong f1-score of 0.9063 both pre- and post-adjudication. The 'Political' frame also performed very well, rising to 0.8867 post-adjudication. On the other hand, 'External Regulation and Reputation' was the most challenging, resulting in a much lower f1-score of 0.6032 due to a low precision of 50.00%.

This human difficulty directly matches the model results, where 'External Regulation' was completely missed by almost all fine-tuned models. However, for trickier frames like 'Capacity and Resources', human discussion during adjudication heavily corrected early mistakes, boosting its f1-score from 0.6670 to 0.7412. This highlights a level of contextual understanding that automated models still cannot match.

5.3 WHAT IS THE INFLUENCE OF HUMANS AND LLMS WORKING TOGETHER ON THE RESULTS OF THE FRAME ANNOTATION TASK OF THE CHANGEMYVIEW DATASET?

To see how humans and LLMs work together, we looked at human performance, inter-annotator agreement, and automation bias across group A and group B.

As shown in table 5.7, the overall f1-score of part B is lower than part A. However, looking at the groups separately, group A improved their f1-score from 0.2592 to 0.2811 when they got LLM suggestions. Group B started with the LLM suggestions and scored 0.2722, but they improved to 0.3444 when they did part A on their own.

Condition	Accuracy	Precision	Recall	F1-score
Overall:				
Part A (No LLM)	0.0850	0.3243	0.3142	0.3018
Part B (With LLM)	0.2250	0.3209	0.2612	0.2778
Group A: Part A First				
Part A (No LLM)	0.0750	0.2733	0.2682	0.2592
Part B (With LLM)	0.2600	0.3387	0.2527	0.2811
Group B: Part B First				
Part A (No LLM)	0.0950	0.4075	0.3601	0.3444
Part B (With LLM)	0.1900	0.3009	0.2697	0.2722

Table 5.7: Performance comparison across conditions with and without LLM suggestions, split by task order.

For both groups, the agreement scores between annotators improved when they got LLM suggestions. As shown in Table 5.8, the overall Jaccard score increased from 0.2436 to 0.3689. Combined with a big increase in overall accuracy (from 0.0850 to 0.2250), this shows that the LLM suggestions successfully guided participants toward the correct gold-standard frames. This helped the annotators agree with each other more and get more exact-match answers right.

Condition	Jaccard	MASI
Overall		
Part A (No LLM)	0.2436	0.1849
Part B (With LLM)	0.3689	0.3112
Group A: Part A First		
Part A (First Task)	0.2294	0.1706
Part B (Second Task)	0.4320	0.3898
Group B: Part B First		
Part A (Second Task)	0.2653	0.2040
Part B (First Task)	0.3200	0.2470

Table 5.8: Inter-annotator agreement measured using Jaccard similarity and MASI across conditions with and without LLM suggestions, split by task order.

For the automation bias, group A followed the LLM more times than group B. This was the case for the correct and incorrect LLM suggestions. In table 5.9 all scores can be found. Group A copied wrong suggestions 21.00% of the time, while group B only copied them 13.00% of the time. A reason for this could be that doing the manual task first fatigued group A, so they were less critical at the LLM in the second part. This could also be the reason why incorrect LLM suggestions dropped the overall human f1-score to 0.1872.

Condition	**Metric**	**Value**
Overall Automation Bias		
Exact Copy Rate (LLM Correct)		0.3050
Exact Copy Rate (LLM Incorrect)		0.1700
Human F1 (Good LLM suggestion)		0.2896
Human F1 (Bad LLM suggestion)		0.1872
Group A: Part A First, Part B Second		
Exact Copy Rate (LLM Correct)		0.3500
Exact Copy Rate (LLM Incorrect)		0.2100
Human F1 (Good LLM suggestion)		0.2979
Human F1 (Bad LLM suggestion)		0.1903
Group B: Part B First, Part A Second		
Exact Copy Rate (LLM Correct)		0.2600
Exact Copy Rate (LLM Incorrect)		0.1300
Human F1 (Good LLM suggestion)		0.2795
Human F1 (Bad LLM suggestion)		0.1826

Table 5.9: Automation bias metrics across conditions, comparing exact copy rates and human F1 scores for correct vs. incorrect LLM suggestions.

To see which frames score best, we look at table A.12. Clear frames like 'Economic' and 'Morality' scored high and stayed stable for both groups. However, difficult frames like 'Quality of Life' showed a big difference. Group A got a score of 0.2800 on their own, but reached a score of 0.7059 when they had LLM suggestions. Frames with a score of zero were most likely because those specific frames did not match any of the comments in that part of the questionnaire.

6 CONCLUSION

"How accurate are LLMs compared to humans for the frame annotation task of the ChangeMyView dataset and what is the influence of humans and LLMs working together on the annotation task?"

This study investigated the accuracy of LLMs compared to human annotators in identifying frames within the CMV dataset and explored the influence of human-LLM collaboration on annotation performance.

As shown in the results, humans remain highly effective in identifying the frames in CMV discussions, achieving reliable f1-scores despite the subjectivity of the task.

Our results show that while LLMs show promise in structured information processing, they often struggle with the nuanced, informal, and emotionally charged language of CMV discussions. These results align with the perspective of Wilkens (2020), confirming that automated systems exhibit clear limitations in linguistic domains that rely on implicit meanings. Consequently, humans remain essential in this task, particularly due to their ability to interpret these emotionally charged languages.

Lastly, this research shows how LLMs work as members of hybrid teams. According to Richter and Richter (2024), hybrid teams should combine human skills with AI to achieve better performance. However, our results show that LLM suggestions did not improve performance. Instead, the overall f1-score dropped because of automation bias.

This shows that trust in a hybrid team is very important. When the LLM makes mistakes, humans often copy them blindly, especially when they are fatigued. Because of this, the higher agreement score is misleading. Humans follow the LLM and agree with each other on the wrong frames. However, group B seem to learn from seeing the LLM errors at the beginning. This taught them not to trust the system blindly and made them more critical later. Since LLMs are not better than humans on the framing task, a hybrid team only works well if humans do not trust the LLM blindly.

6.1 LIMITATIONS AND FUTURE WORK

There are a few limitations to this study. First, since the accuracy of the humans was calculated against their mutually agreed gold standard, it would be interesting to see how other humans would score against already established gold standard. Second, the dataset on which the models are trained and tested is quite small. Therefore, some frames rarely appeared and the models could not learn them well. Third, it is unknown whether the participants in the user study had previous annotation experience.

To explore this further, the models should be trained and tested on a bigger dataset. Also, a new experiment could be done with participants who all have annotation experience to see if they achieve better results with LLM help.

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A APPENDIX

A.1 LINK TO GITHUB

All data, notebooks and python scripts that are used for the analysis in this thesis is available in a public GitHub repository using the following link: https://github.com/anyakiewit/bachelor_thesis

A.2 SAMPLING STATISTICS OF THE 50 DISCUSSIONS

The 50 selected ChangeMyView threads from the Webis-CMV-20 corpus (generated on 2026-04-07) contain the following overall characteristics:

- **Year Balance:** 2013=10, 2014=10, 2015=10, 2016=10, 2017=10 (Total: 50 threads).
- **Delta Distribution (OP changed view):** True=16, False=34.
- **Preserved Comments per Thread:** Min=80, Median=97.0, Mean=94.68, Max=100.
- **Max Reply Depth:** Min=5, Median=10.0, Mean=12.80, Max=53.
- **Unique Authors per Thread:** Min=14, Median=36.5, Mean=38.78, Max=62.
- **Submission Net Upvotes (Score):** Min=0, Median=193.0, Mean=338.08, Max=1271.
- **Deleted/Removed Comments per Thread:** Min=0, Median=3.0, Mean=3.46, Max=12.

All 50 files successfully passed extraction verification, confirming that all required keys, timestamps, and comment trees are present and valid.

A.3 PROMPT FOR SAMPLING 300 COMMENTS

```
"create 3 json files named part_1.json part_2.json part_3.json, each json
must contain 100 different comments and have the same structure as top_200_3.json.
get the different comments from 100_totaal.json and 220_totaal.json.  create
a random collection.  generate the actual json files right away."
```

In this prompt, top_200_3.json is the file containing the first 200 comments from the other project. 100_totaal.json and 220_totaal.json are the files that together contain all 50 discussions. After the LLM generated the three new files with 100 comments each, the output was thoroughly checked to make sure the selections were random and that no formatting issues occurred before starting the annotation phase.

A.4 REFINED ANNOTATION GUIDELINES

Refined Annotation Guidelines - Frames

1. Project Overview

This project aims to identify frames in argumentative texts from the **ChangeMyView (CMV)** dataset. Framing refers to determining which aspect/lens of a topic an argument emphasizes. By identifying the frame, we understand the perspective from which an author is building their argument.

2. Task

- **Task:** For each post or comment, select the most appropriate frame from the list of 15 dimensions.

3. The 15 Framing Dimensions

Use the following definitions based on the *Policy Frames Codebook*:

Economic: costs, benefits, or other financial implications
Capacity and resources: availability of physical, human or financial resources, and capacity of current systems
Morality: religious or ethical implications
Fairness and equality: balance or distribution of rights, responsibilities, and resources
Legality, constitutionality and jurisprudence: rights, freedoms, and authority of individuals, corporations, and government
Policy prescription and evaluation: discussion of specific policies aimed at addressing problems
Crime and punishment: effectiveness and implications of laws and their enforcement
Security and defense: threats to welfare of the individual, community, or nation
Health and safety: health care, sanitation, public safety
Quality of life: threats and opportunities for the individual's wealth, happiness, and well-being
Cultural identity: traditions, customs, or values of a social group in relation to a policy issue
Public opinion: attitudes and opinions of the general public, including polling and demographics
Political: considerations related to politics and politicians, including lobbying, elections, and attempts to sway voters
External regulation and reputation: international reputation or foreign policy of the U.S.
Other: any coherent group of frames not covered by the above categories

Figure 1: Framing dimensions from Boydstun et al. (2014).

1. **Economic:** Costs, benefits, taxes, budgets, financial impact, or efficiency.
2. **Capacity and Resources:** Availability of physical resources, infrastructure, equipment, or manpower.
3. **Morality:** Ethical principles, duty, religious values, or "right vs. wrong" judgments.
4. **Fairness and Equality:** Human/civil rights, discrimination, social justice, or equal distribution of resources.
5. **Constitutionality and Law:** Legal procedures, court cases, the Constitution, or the legality of an action.
6. **Policy Evaluation:** The effectiveness, feasibility, or track record of specific government policies or laws.
7. **Crime and Punishment:** Law enforcement, criminal activity, sentencing, and the justice system.
8. **Security and Defense:** National security, protection against foreign threats, military action, or espionage.
9. **Health and Safety:** Physical health, disease, epidemics, and general safety risks to individuals.
10. **Quality of Life:** Personal well-being, happiness, environmental protection, or daily lifestyle.
11. **Cultural Identity:** National/group traditions, heritage, shared values, or cultural pride.
12. **Public Opinion:** Polls, what "the people" want, popularity, protests, or electoral support.
13. **Political:** Party politics, political strategy, winning/losing power, or election campaigns.

14. **Externalities:** Unintended side effects or indirect consequences (positive or negative) of a decision.
15. **Other:** Only use this for text that is clearly argumentative but does not fit any category above.

→ we used only the 15 frames of Card et al. (2015). We chose to go with these because they provide a comprehensive framework for not only analyzing policy issues but also the informal discussions of the CMV dataset.

4. Annotation Steps (Label Studio)

1. **Read the Original Post (OP):** Understand the main stance of the thread.
2. **Analyze the Sentence:** Identify keywords that signal a perspective (e.g., "tax" → Economic; "unconstitutional" → Law).
3. **Select Label:** Choose the best-fitting frame in **Label Studio**.
4. **Same OP?**: skip the post

4.1 Annotation examples

"*CMV: Religion has no positive attributes*" → morality

"*CMV: Most "big words" have no place outside of formal writing or speeches*." → other

"*CMV: Money to me is a renewable source so I just spend it knowing I'll get paid again so it'll come back*" → economic

"*CMV: There are no adequate gender equal options when it comes to changing your surname in marriage*" → fairness and equality

"*CMV: Watches today serve no functional purpose other than fashion, and therefore are a waste of money*" → public opinion

4.2 Handling Multi-Frame Examples

One OP can have more than one frame. So multi-frame issues are possible. When deciding between two frames, choose both. This is good to reduce subjectivity. For example;

“CMV: I believe that blacks are genetically inferior to whites, Asians, and jews as far as intelligence is concerned.”
→ This could either be ‘morality’ or ‘cultural identity’, so both labels can be chosen.

5. During the annotation process

Doing more annotations, choosing the right frame became easier. This is because by doing the annotation you get a better understanding of the different frames.

after first 150 annotations:
Some frames were still unclear. For example, “Policy prescription and evaluation”. We made those frames clear to each other. Also the frames “Economic” and “Capacity and resources” are both possible in a lot of context, so a lot of times multiframing is possible.

A.5 RESULTS PER ANNOTATOR PRE-ADJUDICATION

Frame	Accuracy	Precision	Recall	f1
Capacity and Resources	0.965	0.8750	0.5385	0.6667
Crime and Punishment	0.985	0.7500	0.8571	0.8000
Cultural Identity	0.940	0.9556	0.8113	0.8776
Economic	0.980	0.9286	0.9286	0.9286
External Regulation and Reputation	0.985	0.5000	0.6667	0.5714
Fairness and Equality	0.955	0.8293	0.9444	0.8831
Health and Safety	0.980	0.8947	0.8947	0.8947
Legality, Constitutionality and Jurisprudence	0.950	0.9091	0.7143	0.8000
Morality	0.950	0.8571	0.9818	0.9153
Other	0.960	0.7308	0.9500	0.8261
Policy Prescription and Evaluation	0.950	0.7692	0.8333	0.8000
Political	0.985	0.8929	1.0000	0.9434
Public Opinion	0.945	0.7619	0.7273	0.7442
Quality of Life	0.955	0.8485	0.8750	0.8615
Security and Defense	0.995	0.8750	1.0000	0.9333
Total for all frames	0.725	0.8625	0.8879	0.8658

Table A.1: Performance of Annotator 1 pre-adjudication.

Frame	Accuracy	Precision	Recall	f1
Capacity and Resources	0.965	0.6667	0.9231	0.7742
Crime and Punishment	0.985	0.7000	1.0000	0.8235
Cultural Identity	0.885	0.7419	0.8679	0.8000
Economic	0.980	0.8750	1.0000	0.9333
External Regulation and Reputation	0.985	0.5000	0.6667	0.5714
Fairness and Equality	0.935	0.8485	0.7778	0.8116
Health and Safety	0.940	0.6522	0.7895	0.7143
Legality, Constitutionality and Jurisprudence	0.965	0.8387	0.9286	0.8814
Morality	0.925	0.8571	0.8727	0.8649
Other	0.975	1.0000	0.7500	0.8571
Policy Prescription and Evaluation	0.945	0.8824	0.6250	0.7317
Political	0.975	0.9545	0.8400	0.8936
Public Opinion	0.965	0.7778	0.9545	0.8571
Quality of Life	0.910	0.7917	0.5938	0.6786
Security and Defense	0.970	0.5455	0.8571	0.6667
Total for all frames	0.560	0.8192	0.8408	0.8092

Table A.2: Performance of Annotator 2 pre-adjudication.

Frame	Accuracy	Precision	Recall	f1
Capacity and Resources	0.945	0.5833	0.5385	0.5600
Crime and Punishment	0.990	0.7778	1.0000	0.8750
Cultural Identity	0.895	0.7581	0.8868	0.8174
Economic	0.965	1.0000	0.7500	0.8571
External Regulation and Reputation	0.985	0.5000	1.0000	0.6667
Fairness and Equality	0.895	0.6667	0.8333	0.7407
Health and Safety	0.970	0.8421	0.8421	0.8421
Legality, Constitutionality and Jurisprudence	0.950	0.8462	0.7857	0.8148
Morality	0.895	0.8696	0.7273	0.7921
Other	0.930	0.6364	0.7000	0.6667
Policy Prescription and Evaluation	0.905	0.5714	0.8333	0.6780
Political	0.950	0.8947	0.6800	0.7727
Public Opinion	0.910	0.5714	0.7273	0.6400
Quality of Life	0.925	0.7429	0.8125	0.7761
Security and Defense	0.985	1.0000	0.5714	0.7273
Total for all frames	0.520	0.7429	0.7796	0.7458

Table A.3: Performance of Annotator 4 pre-adjudication.

A.6 DUPLICATE COMMENTS

ID	Task info	Annotator	Thread
ch26fk7	triple 20874 / single 21451	4	CMV: I believe that current political campaign norms are wasteful
dcftrwo	triple 20830 / single 21299	1	CMV: I can't see why I should have to respect Christianity
deywptq	triple 20949 / single 21258	1	CMV: The United States can and should act as the "World Police"
deyyihc	triple 20911 / single 21428	4	Duplicate of 3: US as "World Police"
djeqp2j	triple 20862 / single 21222	1	CMV: Personal problems do not justify being rude to others

Table A.4: The duplicate comments found in the dataset.

A.7 RESULTS PER ANNOTATOR POST-ADJUDICATION

Frame	**Accuracy**	**Precision**	**Recall**	**f1**
Capacity and Resources	0.975	0.9000	0.6923	0.7826
Crime and Punishment	0.985	0.7500	0.8571	0.8000
Cultural Identity	0.940	0.9556	0.8113	0.8776
Economic	0.980	0.9286	0.9286	0.9286
External Regulation and Reputation	0.985	0.5000	0.6667	0.5714
Fairness and Equality	0.960	0.8333	0.9722	0.8974
Health and Safety	0.980	0.8947	0.8947	0.8947
Legality, Constitutionality and Jurisprudence	0.950	0.9091	0.7143	0.8000
Morality	0.950	0.8571	0.9818	0.9153
Other	0.970	0.7917	0.9500	0.8636
Policy Prescription and Evaluation	0.955	0.7778	0.8750	0.8235
Political	0.985	0.8929	1.0000	0.9434
Public Opinion	0.955	0.7826	0.8182	0.8000
Quality of Life	0.960	0.8529	0.9062	0.8788
Security and Defense	0.995	0.8750	1.0000	0.9333
Total for all frames	0.745	0.8725	0.9029	0.8791

Table A.5: Performance of Annotator 1 post-adjudication.

Frame	Accuracy	Precision	Recall	f1
Capacity and Resources	0.965	0.6667	0.9231	0.7742
Crime and Punishment	0.985	0.7000	1.0000	0.8235
Cultural Identity	0.890	0.7541	0.8679	0.8070
Economic	0.980	0.8750	1.0000	0.9333
External Regulation and Reputation	0.985	0.5000	0.6667	0.5714
Fairness and Equality	0.940	0.8529	0.8056	0.8286
Health and Safety	0.940	0.6522	0.7895	0.7143
Legality, Constitutionality and Jurisprudence	0.965	0.8387	0.9286	0.8814
Morality	0.930	0.8596	0.8909	0.8750
Other	0.975	1.0000	0.7500	0.8571
Policy Prescription and Evaluation	0.950	0.8889	0.6667	0.7619
Political	0.980	0.9565	0.8800	0.9167
Public Opinion	0.965	0.7778	0.9545	0.8571
Quality of Life	0.920	0.8077	0.6562	0.7241
Security and Defense	0.970	0.5455	0.8571	0.6667
Total for all frames	0.580	0.8217	0.8521	0.8181

Table A.6: Performance of Annotator 2 post-adjudication.

Frame	Accuracy	Precision	Recall	f1
Capacity and Resources	0.955	0.6429	0.6923	0.6667
Crime and Punishment	0.990	0.7778	1.0000	0.8750
Cultural Identity	0.900	0.7705	0.8868	0.8246
Economic	0.965	1.0000	0.7500	0.8571
External Regulation and Reputation	0.985	0.5000	1.0000	0.6667
Fairness and Equality	0.900	0.6818	0.8333	0.7500
Health and Safety	0.970	0.8421	0.8421	0.8421
Legality, Constitutionality and Jurisprudence	0.950	0.8462	0.7857	0.8148
Morality	0.900	0.8723	0.7455	0.8039
Other	0.935	0.6667	0.7000	0.6829
Policy Prescription and Evaluation	0.910	0.5882	0.8333	0.6897
Political	0.955	0.9000	0.7200	0.8000
Public Opinion	0.920	0.6000	0.8182	0.6923
Quality of Life	0.935	0.7568	0.8750	0.8116
Security and Defense	0.985	1.0000	0.5714	0.7273
Total for all frames	0.540	0.7554	0.7958	0.7608

Table A.7: Performance of Annotator 4 post-adjudication.

A.8 PROMPT ENGINEERING

Prompt Used for LLM Frame Annotation

You are an expert annotator for framing analysis in argumentative discourse, using the Boydstun et al. (2014) media frame taxonomy.

You will be given:

- The original post (OP) of an r/ChangeMyView thread — context only.
- A single comment replying to the OP — the target for annotation.

For the comment (not the OP), identify which frames are present from the taxonomy below.

Frame Definitions

- **Economic**: Costs, benefits, or other financial implications.
- **Capacity and Resources**: Availability of physical, human, or financial resources, and system capacity.
- **Morality**: Religious or ethical implications.
- **Fairness and Equality**: Distribution of rights, responsibilities, and resources.
- **Legality, Constitutionality and Jurisprudence**: Rights, freedoms, and authority of individuals, corporations, and government, including constitutional or legal interpretation.
- **Policy Prescription and Evaluation**: Discussion of specific policies aimed at addressing or evaluating problems.
- **Crime and Punishment**: Effectiveness, fairness, and implications of laws and their enforcement.
- **Security and Defense**: Threats to the safety of individuals, communities, or nations.
- **Health and Safety**: Issues related to healthcare, sanitation, and public safety.
- **Quality of Life**: Impacts on individual wealth, happiness, well-being, or life satisfaction.
- **Cultural Identity**: Traditions, customs, or values of social groups in relation to the issue.
- **Public Opinion**: Attitudes, beliefs, or polling of the general public.
- **Political**: Political actors, elections, lobbying, or strategic political considerations.
- **External Regulation and Reputation**: International relations, foreign policy, or global reputation.
- **Other**: Any coherent framing not captured by the above categories.

Rules

- A comment may include multiple frames (multi-label classification is allowed).
- Only assign a frame if it is substantively present, not if it is merely mentioned.
- Return JSON with exactly one key: `"comment_frames"`.
- The value must be a list of frame labels using the exact names provided above.
- If no frames apply, return an empty list.

Input
Title: `{thread_title}`
Original Post (context only):
`{thread_body}`
Comment to annotate:
`{comment_body}`

A.9 ANNOTATION GUIDELINE QUESTIONNAIRE

Please read each comment carefully and determine which "frame" the author is using to make their point.

Rules

- Select 1 or 2 frames: If an argument clearly touches on two perspectives, please select both.
- However, when choosing 'other', only 1 label is allowed
- Be Critical: In part two you will see suggestions provided by an AI. Please use your own judgment to decide if the AI is correct or if a different frame fits better.

Examples

- "CMV: Religion has no positive attributes." → *Morality*
- "CMV: Most 'big words' have no place outside of formal writing or speeches." → *Other*
- "CMV: Money to me is a renewable source so I just spend it knowing I'll get paid again so it'll come back."
  → *Economic*
- "CMV: There are no adequate gender-equal options when it comes to changing your surname in marriage."
  → *Fairness and Equality*
- "CMV: I believe that blacks are genetically inferior to whites, Asians, and Jews as far as intelligence is concerned."
  → This could either be *Morality* or *Cultural Identity* , so both labels can be chosen.

The frames:

You will choose from the 15 frames defined Card et al. (2015), see (the table) below. The frames and a brief description of each frame will be provided with every question to help you choose.

- **Economic —** Costs, benefits, or other financial implications.
- **Capacity and Resources —** Availability of physical, human, or financial resources, and capacity of current systems.
- **Morality —** Religious or ethical implications.
- **Fairness and Equality —** Balance or distribution of rights, responsibilities, and resources.
- **Legality, Constitutionality and Jurisprudence —** Rights, freedoms, and authority of individuals, corporations, and government.
- **Policy Prescription and Evaluation —** Discussion of specific policies aimed at addressing problems.
- **Crime and Punishment —** Effectiveness and implications of laws and their enforcement.
- **Security and Defense —** Threats to the welfare of the individual, community, or nation.

- **Health and Safety** — Health care, sanitation, and public safety.
- **Quality of Life** — Threats and opportunities for an individual's wealth, happiness, and well-being.
- **Cultural Identity** — Traditions, customs, or values of a social group in relation to a policy issue.
- **Public Opinion** — Attitudes and opinions of the general public, including polling and demographics.
- **Political** — Considerations related to politics and politicians, including lobbying, elections, and attempts to sway voters.
- **External Regulation and Reputation** — International reputation or foreign policy of the United States.
- **Other** — Any coherent group of frames not covered by the above categories.

A.10 QUESTIONNAIRE

Table A.8: Comments and gold frames of Part A of the forms.

#	Gold Frames	Comment
1	morality + fairness and equality	Children deserve no respect because they cant hurt you. Once they are big enough, you can respect them
2	cultural identity	I would mind and FEAR if they acted like us on THEIR OWN.
3	external regulation and reputation	It's a geographical thing – westerners have no right to organize protests on the domestic policies of the easterners, just as easterners have no right to criticize the west for its domestic policies.
4	fairness and equality + crime and punishment	Aren't public officials the only people that CAN engage in corruption? How do you make all corruption sentences stronger than the average corruption sentence?
5	morality + public opinion	If people can realize they're OK with some GMOs outside the food chain, they might think differently about the GMOs they are against inside the food chain.
6	morality + cultural identity	I don't know if it matters to you at all (I bring it up because many politicians claim Christianity), but the Bible pretty much teaches what you're saying. Those in leadership positions should be held to a higher standard and be beyond reproach.
7	capacity and resources	In many countries any commercial content in schools is forbidden (e.g. Israel).
8	other	When that person shoots you because no one cared about or understood his problems/frustration, maybe that will change your mind.
9	political + public opinion	The fact that Hillary stole the primary from Bernie left a lot of rightfully salty Bernie supporters. Many stuck with Clinton for the general, some wrote in Bernie, some probably didn't vote, some went third, and some probably went over to Trump, would you believe.
10	policy prescription and evaluation	Civil forfeiture has done some good. Forfeiture programs "have crippled powerful drug-trafficking organizations".
11	quality of life + cultural identity	Naturally letting society evolve to tolerate differences will almost undoubtedly lead to a higher tolerance of those differences than having the state enforce tolerance.
12	other	The GOP has no choice. The roles forgot the PSU from kicking the nominee out at this point. QED.
13	quality of life + capacity and resources	I think the most realistic outcome of the development of strong AI is not a black and white scenario where there are the big scary evil "Robots" and the weakened soft fleshy "Humans" are forever separate; instead, surely we'll slowly merge with robots / robots will slowly merge with us.

#	Gold Frames	Comment
14	economic + fairness and equality	An important thing to realize is that these issues come from the economic system. Let's look at capitalism and sexism. When sexism is taught and shown all around us, it is much easier to pay women less than men, increasing profit. Then when they are stay-at-home baby makers, they can produce workers and make sure they can work in 20 years and be better with mommy's 100% attention. Sexism is profitable.
15	quality of life + public opinion	Why would a larger audience for this deception make the world a better place and not a more confused place?
16	morality	To some people, the base meaning of respect is to be treated like a person. To others, the base meaning of respect is to be treated like an authority.
17	crime and punishment + public opinion	You would advocate the use of force to stop someone from tipping? That is to say, you would put someone in jail if they tipped someone else? What about the person who accepts that tip? Is he an accomplice to the crime?
18	economic	Children seem like a horribly inefficient retirement plan. If you are only concerned about the financial aspects, it would be way cheaper to take the money you would have spent raising children, put it in an IRA invested in an index fund, and you'll come out in pretty good shape financially when retirement rolls around.
19	health and safety + policy prescription	It says non-FDA-approved drugs; it specifically mentions that they should be currently undergoing trials.
20	morality	Now let's imagine it's "human" – like us, as you say. It would feel power, greed, etc., the nasty stuff. But it would also feel empathy, and in fact that is what an AI needs to not become evil.

Table A.9: Comments, gold frames, LLM suggestions of Part B of the forms.

#	Right Frames	LLM Suggestion	Correct/incorrect suggestion	Comment
21	quality of life	quality of life	correct	You really think out of billions of people there aren't people who enjoy and build a strong relationship out of the game of "winning one another over" though?
22	morality + public opinion	economic	incorrect	No you're wrong. If the media gave us the facts and told the truth instead of throwing their worthless opinions in we could watch more reliable television.
23	other	other	correct	Being still and alert for hours at a time, that upper level eSport participants do, is a physical exertion. It's physical because it requires sitting in one spot and being alert. It's an exertion because you are sore and tired at the end of it.
24	morality	external regulation	incorrect	He's your answer: respect everyone unless proven otherwise.

#	Gold Frames	LLM Suggestion	Correct/incorrect suggestion	Comment
25	Legality, constitutionality and jurisprudence + morality	Legality, constitutionality and jurisprudence + morality	correct	What if the only way to "legally" obtain a movie is to pay $1,000,000 dollars for it? Would it be immoral to pirate that?
26	economic + fairness and equality	capacity and resources	incorrect	Gender inequality **is** a huge economic issue for half the population who are underpaid for their work or discriminated against for promotion etc.
27	economic + quality of life	economic + quality of life	correct	Every child is a retirement plan. I hope I raise my kids to make a better world. A world that will make my elderly ass easier.
28	fairness and equality	fairness and equality	correct	Do you see the problem here? It is intentionally obtuse to argue that diversity is unrelated to discrimination. Problems with discrimination are what give rise to the appeal of diversity in the first place.
29	crime and punishment	public opinion	incorrect	As for reason 1, I would argue that death, even at your own choosing, is the ultimate punishment.
30	Legality, constitutionality and jurisprudence + fairness and equality	Legality, constitutionality and jurisprudence + fairness and equality	correct	Sites stealing Reddit content?! Reddit content is stolen off other sites too – not to mention 99% of images are posted to Imgur, lacking any attribution to the original creator, much less the site they were taken from.
31	security and defense + morality	security and defense + morality	correct	What I've said is precisely what I'm saying: that if gay pride parades are part of a strategy to improve gay rights, they are, at this time, doing more harm than good.
32	external regulation and reputation	health and safety	incorrect	The United States, being the only nation on earth that currently has the military capacity to destroy any other regime on earth, has a duty and a responsibility to function as a de-facto police force for the world.

#	Gold Frames	LLM Suggestion	Correct/incorrect suggestion	Comment
33	economic + crime and punishment	health and safety + policy prescription	correct	Why should it be illegal to give someone a small sum of money for good service? Who are you to tell an entire nation that practice is not to be allowed?
34	morality	political	incorrect	Yeah, and my contention is that your view is not only different than mine, but morally wrong.
35	economic	economic	correct	Advertisements should be allowed in school because it is an easy way to generate revenue and as you noted there would be high demand to advertise in schools since kids are such an important demographic for marketers.
36	economic + policy prescription and evaluation	morality	incorrect	I believe that you are missing an enormous part of campaigning with your proposed reforms, namely the *soft money* campaigning of corporations, unions, and the more wealthy members of the US.
37	Legality, constitutionality and jurisprudence + policy evaluation	Legality, constitutionality and jurisprudence + policy evaluation	correct	That being said, expanding that judgment to people with clean records and a single traffic violation in question is clearly a case of a poorly written rule set.
38	fairness and equality + quality of life	security and defense	incorrect	I have spoken to Glaswegians, Californians, Texans, Aussies, Kiwis, Corkmen, Dubliners etc. and they've *all* spoken the same language. Sometimes they have slight differences in word preference, and they often have different slang. They always have significant differences in pronunciation.
39	Legality, constitutionality and jurisprudence	capacity and resources	incorrect	When you steal, you don't gain ownership of whatever you stole, you merely have it in your possession. The person from whom it was stolen still has ownership.

#	Gold Frames	LLM Suggestion	Correct/incorrect suggestion	Comment
40	other	public opinion	incorrect	Even Reuters and AP are inherently editorialized through omission, selection, language, geography, and so forth. You can't really have "Just the facts" unless you have all the facts.

A.11 MEAN OF RESULTS PER ANNOTATOR (PRE- AND POST-ADJUDICATION

Frame	Accuracy	Precision	Recall	f1
Capacity and Resources	0.9583	0.7083	0.6667	0.6670
Crime and Punishment	0.9867	0.7426	0.9524	0.8328
Cultural Identity	0.9067	0.8185	0.8553	0.8317
Economic	0.9750	0.9345	0.8929	0.9063
External Regulation and Reputation	0.9850	0.5000	0.7778	0.6032
Fairness and Equality	0.9283	0.7815	0.8518	0.8118
Health and Safety	0.9633	0.7963	0.8421	0.8170
Legality, Constitutionality and Jurisprudence	0.9550	0.8647	0.8095	0.8321
Morality	0.9233	0.8613	0.8606	0.8574
Other	0.9550	0.7891	0.8000	0.7833
Policy Prescription and Evaluation	0.9333	0.7410	0.7639	0.7366
Political	0.9700	0.9140	0.8400	0.8699
Public Opinion	0.9400	0.7037	0.8030	0.7471
Quality of Life	0.9300	0.7944	0.7604	0.7721
Security and Defense	0.9833	0.8068	0.8095	0.7758
TOTAL (Mean of Annotators)	0.6017	0.8082	0.8361	0.8069

Table A.10: Mean performance across annotators by frame (pre-adjudication).

Frame	Accuracy	Precision	Recall	f1
Capacity and Resources	0.9650	0.7365	0.7692	0.7412
Crime and Punishment	0.9867	0.7426	0.9524	0.8328
Cultural Identity	0.9100	0.8267	0.8553	0.8364
Economic	0.9750	0.9345	0.8929	0.9063
External Regulation and Reputation	0.9850	0.5000	0.7778	0.6032
Fairness and Equality	0.9333	0.7893	0.8704	0.8253
Health and Safety	0.9633	0.7963	0.8421	0.8170
Legality, Constitutionality and Jurisprudence	0.9550	0.8647	0.8095	0.8321
Morality	0.9267	0.8630	0.8727	0.8647
Other	0.9600	0.8195	0.8000	0.8012
Policy Prescription and Evaluation	0.9383	0.7516	0.7917	0.7584
Political	0.9733	0.9165	0.8667	0.8867
Public Opinion	0.9467	0.7201	0.8636	0.7831
Quality of Life	0.9383	0.8058	0.8125	0.8048
Security and Defense	0.9833	0.8068	0.8095	0.7758
TOTAL (Mean of Annotators)	0.6217	0.8165	0.8503	0.8193

Table A.11: Mean performance across annotators by frame (post-adjudication).

A.12 PER-FRAME F1 SCORES ACROSS GROUPS

Category	Overall	Part A	Part B
Group A			
economic	0.6667	0.7895	0.6053
capacity/resources	0.0000	0.0000	0.0000
morality	0.5285	0.4632	0.5918
fairness/equality	0.5210	0.4231	0.5970
law/constitution	0.6022	0.0000	0.7568
policy eval	0.1370	0.1081	0.1667
crime/punishment	0.2647	0.2927	0.2222
security/defense	0.0000	0.0000	0.0000
health/safety	0.1290	0.1667	0.0000
quality of life	0.4950	0.2800	0.7059
cultural identity	0.2222	0.2593	0.0000
public opinion	0.2174	0.3175	0.0000
political	0.4865	0.5455	0.0000
external reg.	0.2353	0.1250	0.3333
other	0.1818	0.1176	0.2373
Group B			
economic	0.6306	0.8718	0.5000
capacity/resources	0.1364	0.2222	0.0000
morality	0.5340	0.5287	0.5385
fairness/equality	0.3385	0.2182	0.4267
law/constitution	0.5155	0.0000	0.6579
policy eval	0.3678	0.3721	0.3636
crime/punishment	0.3492	0.5556	0.0741
security/defense	0.0741	0.0000	0.1053
health/safety	0.3243	0.3750	0.0000
quality of life	0.4854	0.3922	0.5769
cultural identity	0.2326	0.3125	0.0000
public opinion	0.1875	0.2373	0.1081
political	0.3404	0.4706	0.0000
external reg.	0.3871	0.3333	0.4211
other	0.2931	0.2759	0.3103

Table A.12: Per-frame F1 scores across conditions with and without LLM suggestions, split by task order.